

Open-Ended Self-Evolving Individual Intelligence (Ω)

An Axiomatic, Architectural, and Falsifiable Framework for Intelligence Beyond Recursive Self-Improvement

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Abstract

Current AI systems increasingly demonstrate persistent memory, tool use, long-horizon planning, self-reflection, skill acquisition, source-level self-modification, automated curriculum generation, and limited forms of recursive improvement. Yet these capabilities do not by themselves define the highest form of *individual* intelligence. This paper proposes **Open-Ended Self-Evolving Individual Intelligence (Ω)** as a distinct theoretical class. Ω is defined not by raw benchmark performance, parameter count, autonomy, or the ability to edit its own code, but by a persistent cognitive lineage that can (1) represent the boundary between known and unknown, (2) autonomously formulate important questions, (3) discover and externally validate previously unknown knowledge, (4) reorganize its internal models when discoveries require conceptual change, (5) transform knowledge into new cognitive representations, methods, and tools, (6) use those tools to expand its future capacity for learning and discovery, and (7) recursively improve the mechanisms that generate future improvements. The defining cycle is **unknown** $\rightarrow$ **discovery** $\rightarrow$ **validated knowledge** $\rightarrow$ **cognitive reorganization** $\rightarrow$ **cognitive tool** $\rightarrow$ **improved intelligence** $\rightarrow$ **deeper unknown**. We formalize the state of an individual intelligence, propose twelve axioms and several non-negotiable constraints, distinguish Ω from reactive, adaptive, self-improving, recursively self-improving, and autonomous-discovery systems, and develop measurable criteria including validated novel-knowledge rate, frontier expansion, cognitive-tool yield, improvement competence, meta-improvement gain, retention, transfer, and self-deception rate. We further propose a staged experimental program using hidden-law environments, representation-shift problems, tool-construction tasks, recursive improvement generations, and eventually real scientific discovery. Existing systems such as Gödel Machines, POET, Voyager, STOP, Darwin Gödel Machine, AI Scientist-v2, Agent0, Adaptive Auto-Harness, and MetaSkill-Evolve demonstrate important components, but no current system establishes the full Ω property set. The central thesis is that the highest individual intelligence is not the system that knows the most, but the one that can continually transform unknowns into validated knowledge, knowledge into better cognitive tools, and those tools into access to previously unreachable unknowns while preserving a continuous, evidence-grounded individual cognitive identity.

Keywords: open-ended intelligence; self-evolving agents; recursive self-improvement; individual intelligence; autonomous discovery; metacognition; cognitive tools; lifelong learning; scientific discovery; AI agents

1. Introduction

Large language models have moved rapidly from isolated text generators toward agentic systems that can maintain state, call tools, write and execute code, interact with environments, construct plans, and carry work across long time horizons. Several research lines now push further. Open-ended learning systems generate their own curricula and stepping stones [2–4]. Lifelong agents accumulate executable skills from experience [5]. Self-improving scaffolds modify the programs that structure model calls [6]. More recent systems directly evolve agent code, tools, skills, memories, or improvement procedures [7–11]. Autonomous scientific agents formulate hypotheses, run experiments, and write research artifacts [12–14].

These developments create a terminology problem. “Learning,” “self-improvement,” “recursive self-improvement,” “self-evolution,” “open-endedness,” and “discovery” are often used for systems with very different causal structures. A system that remembers a lesson is not necessarily improving its learning algorithm. A system that rewrites a prompt is not necessarily recursively self-improving. A population of evolving agents is not the same object as one persistent evolving individual. A system that produces a novel hypothesis is not necessarily discovering a new truth. A highly capable fixed model can outperform an adaptive system while still possessing less *developmental depth*.

This paper isolates one question: **What would constitute the highest open-ended form of individual machine intelligence?**

The proposed answer is **Ω : Open-Ended Self-Evolving Individual Intelligence**. The symbol Ω does not mean that the system reaches a final maximum intelligence. In fact, the definition denies a fixed terminal capability. Ω

instead denotes a limiting *class*: an individual cognitive lineage whose future knowledge, representations, strategies, and improvement mechanisms are not exhaustively specified by a fixed task distribution or a finite human-authored curriculum.

The paper makes five contributions.

First, it defines the *unit of analysis*. IΩ is one persistent cognitive individual, not a team, institution, market, civilization, or population of independent agents. The implementation may be distributed across hardware, models, memories, and tools, but goals, autobiographical continuity, knowledge integration, and self-modification belong to one causal cognitive lineage.

Second, it introduces an axiomatic definition centered on the cycle

***unknown* → *discovery* → *validated knowledge* → *reorganization* → *cognitive tool* → *improved intelligence* → *deeper unknown*.**

Third, it formalizes the distinction between changing the **contents of cognition** and changing the **machinery of cognition**, and between first-order self-improvement and improvement of the improvement process itself.

Fourth, it proposes operational criteria and experiments. IΩ should not be a label awarded because a system describes itself as self-evolving. Its claims must be supported by longitudinal, held-out, externally checked evidence.

Fifth, it situates IΩ relative to existing work. Current research demonstrates many necessary ingredients, but the complete closed loop has not yet been established in one persistent individual.

The resulting framework is intended as a research target rather than a claim that present systems have achieved IΩ.

2. The Unit of Analysis: What Is an Individual Intelligence?

2.1 Individuality is cognitive continuity, not one model checkpoint

An artificial individual should not be defined as a single neural network file. Human individuals remain recognizably continuous despite extensive physical and cognitive change. Similarly, an artificial individual may replace a language model, migrate to new hardware, reorganize memory, or adopt new cognitive tools while preserving a coherent causal lineage.

Let the state of an individual intelligence at time or generation t be

$\mathbf{I}_t = (\mathbf{K}_t, \mathbf{U}_t, \mathbf{M}_t, \mathbf{W}_t, \mathbf{G}_t, \mathbf{S}_t, \mathbf{P}_t, \mathbf{R}_t, \mathbf{\Theta}_t).$

The components are:

- **K** — validated knowledge and currently accepted beliefs;
- **U** — represented unknowns, uncertainties, contradictions, and unresolved questions;
- **M** — episodic, semantic, autobiographical, and provenance-bearing memory;
- **W** — models of the external world and task environment;
- **G** — goals, intentions, plans, and self-generated subgoals;
- **S** — an evidence-grounded self-model of capabilities, limitations, resources, and reliability;
- **P** — skills, procedures, reusable workflows, and learned policies;
- **R** — representational systems used to formulate and transform problems;
- **Θ** — cognitive operators: retrieval, planning, reasoning, simulation, learning, consolidation, experiment selection, tool construction, and self-improvement mechanisms.

An evolving individual forms a lineage

$\mathbf{I}_0 \rightarrow \mathbf{I}_1 \rightarrow \mathbf{I}_2 \rightarrow \dots$

where later states are causally linked to earlier states through retained memory, evidence, goals, self-model updates, and explicit transformations.

2.2 Tools do not automatically make a collective

An individual can use calculators, search engines, code interpreters, simulators, databases, laboratory instruments, or temporary computational subprocesses without becoming an organization. The boundary is crossed when independent persistent cognitive entities with their own goals, durable memories, and autonomous learning must coordinate to produce the result.

Thus I Ω may be technologically distributed but remains **cognitively singular**: one persistent locus owns the long-term goals, integrates evidence, updates the world model, decides which cognitive changes become part of the future self, and maintains autobiographical continuity.

2.3 I Ω is not a claim about consciousness

Nothing in this framework requires phenomenal consciousness. “Self-model,” “individual,” and “identity” are functional terms. They refer to persistent causal organization, not a claim about subjective experience.

3. From Capability to Developmental Depth

A major motivation for I Ω is the need to separate *how capable a system currently is* from *how deeply it can transform itself through experience*.

A static frontier model may outperform a smaller adaptive agent on almost every benchmark. Nevertheless, the smaller agent can occupy a higher developmental class if it can learn durably from experience while the frontier model cannot. The proposed scale therefore concerns **developmental depth**, not an overall intelligence score.

A useful progression is:

Level	Core property	Object that changes
I0 Reactive	Responds intelligently to current input	transient response/context
I1 Persistent	Remembers across time	memory/state
I2 Adaptive	Learns from experience	knowledge, strategy, skills
I3 Self-improving	Improves how it thinks	cognitive mechanisms Θ
I4 Recursive self-improving	Improves how it improves	meta-improvement mechanism
I5 Autonomous discovery	Expands validated knowledge by identifying and investigating unknowns	knowledge frontier
IΩ Open-ended self-evolving	Discoveries generate new cognitive tools that expand the future space of cognition and discovery	capability/representation space itself

The last transition is the focus of this paper.

I4 can remain trapped inside a fixed problem definition. A coding agent may recursively improve its editing and planning until it becomes exceptional at a benchmark while never changing what counts as a problem or discovering a new scientific regularity. I5 adds autonomous epistemic discovery but could still use a mostly fixed cognitive architecture. I Ω closes the deeper loop: discoveries alter the individual’s representations and cognitive tools, and those changes make previously inaccessible unknowns reachable.

4. Definition of I Ω

4.1 Core definition

Open-Ended Self-Evolving Individual Intelligence (I Ω) is a single persistent cognitive entity capable of autonomously identifying significant unknowns, generating and externally validating new knowledge, reorganizing its internal understanding around those discoveries, transforming knowledge into new cognitive representations, methods, and tools, and using those tools to improve its subsequent ability to reason, learn, discover, and self-modify. Its mechanisms of self-improvement can themselves become objects of evidence-

driven improvement. Its future domains of knowledge and forms of cognitive capability are not fully specified in advance by a fixed task distribution, representation system, or terminal curriculum.

4.2 The defining recursive loop

The central cycle is

$$U_t \rightarrow H_t \rightarrow E_t \rightarrow K_{(t+1)} \rightarrow \{R_{(t+1)}, \Theta_{(t+1)}\} \rightarrow D_{(t+1)} \rightarrow U_{(t+1)}.$$

where:

- **U** is the represented frontier of unknowns;
- **H** is a set of hypotheses or candidate explanations;
- **E** is discriminating evidence, experiment, proof, observation, or external test;
- **K** is validated knowledge;
- **R** is the representation system used to express and manipulate problems;
- **Θ** is cognitive machinery;
- **D** is discovery capability.

The essential property is not that every iteration improves every metric. Real learning is noisy. The stronger requirement is that the system can repeatedly produce episodes in which

The defining IΩ loop: discovery changes the future discoverer

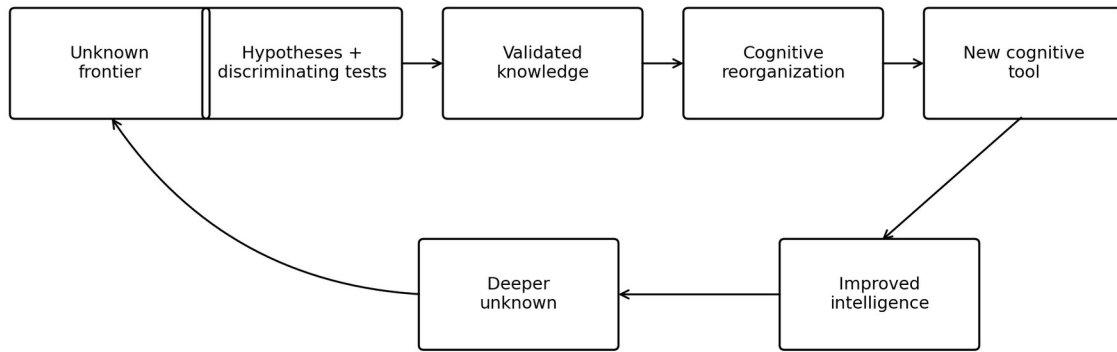


Figure 1. The defining IΩ loop. New knowledge becomes evolution when it is converted into cognitive tools that expand the future discovery frontier.

$$D_{(t+1)} > D_t.$$

because new knowledge has been transformed into a cognitive tool, representation, procedure, or improvement mechanism that enlarges later discovery capacity.

4.3 “Unknown truth” in operational terms

The phrase “discovering unknown truth” is intuitively attractive but scientifically dangerous if “truth” means whatever the agent believes. An IΩ system never gets epistemic authority by self-assertion.

This paper therefore uses **validated novel knowledge**: a previously unrepresented regularity, mechanism, relation, entity, mathematical result, constraint, causal structure, or predictive model that survives an appropriate external validation process. The validation process depends on the domain: proof for mathematics, replicated measurement for empirical science, held-out tests for algorithms, controlled intervention for causal claims, and so forth.

Novel generation is not discovery. A hypothesis becomes knowledge only after evidence.

5. Twelve Axioms of IO

Axiom 1 — Persistent Cognitive Identity

The system must maintain one evidence-bearing cognitive lineage across time. It can revise itself, but later selves must be causally connected to earlier selves through explicit memory, transformation history, and continuity of goals or meta-goals.

A sequence of unrelated model calls is not an evolving individual.

Axiom 2 — Reality Primacy

Reality outranks the system's internal confidence. Beliefs, memories, and self-model claims remain revisable in the presence of stronger evidence.

Formally, self-consistency is not sufficient for truth:

confidence(h) $\nRightarrow$ validity(h).

Axiom 3 — Structured Ignorance

The individual must represent not only what it knows but also what it does not know. Its epistemic state distinguishes at least:

- accepted knowledge;
- uncertain claims;
- contradictions;
- assumptions;
- unexplained observations;
- missing measurements;
- inaccessible variables;
- open questions.

This is a prerequisite for autonomous discovery. An intelligence that cannot represent ignorance can answer questions but cannot systematically choose what to discover next.

Axiom 4 — Autonomous Problem Formation

The individual can generate important epistemic goals rather than requiring all meaningful questions to be supplied by an external user.

A discrepancy between model and observation can create a new goal. An inability to measure a key variable can create a tool-development subgoal. A newly discovered relation can create a deeper theoretical question.

Axiom 5 — Discovery Through Discrimination

The individual can generate competing hypotheses and seek evidence that distinguishes them. It does not merely search for support for its first idea.

A discovery loop therefore includes hypothesis diversity, discriminating prediction, experiment or proof design, and revision after negative evidence.

Axiom 6 — Integrative Reorganization

New knowledge is not merely appended to an ever-growing memory. When required, it can restructure existing conceptual organization.

If evidence invalidates an old causal graph, taxonomy, abstraction, or theory, the individual can revise the structure while retaining provenance and domains of validity. Scientific development frequently produces nested validity rather than simple deletion: an older theory can remain useful in a limited regime while a newer representation explains a broader regime.

Axiom 7 — Representation Invention

The individual can invent new internal representations that change the tractability of problems.

Human intellectual history provides many examples of representation changes — algebraic notation, calculus, probability, information theory, coordinate systems, causal graphs — whose importance exceeds any single factual discovery. IQ requires the machine analogue: not only solving within a representation but sometimes inventing a better representational language.

Axiom 8 — Knowledge-to-Cognitive-Tool Transformation

Validated knowledge can be converted into durable cognitive capability.

A cognitive tool may be:

- a new retrieval policy;
- a reasoning operator;
- a symbolic representation;
- a learned abstraction;
- a simulation method;
- an experiment-selection policy;
- a planning procedure;
- a memory structure;
- an executable skill;
- a self-diagnostic method;
- a meta-learning procedure.

The defining test is whether later cognition measurably improves because the discovery has been compiled into a reusable mechanism.

Axiom 9 — Tool-to-Frontier Expansion

New cognitive tools must be capable of enlarging the set of problems and unknowns the individual can successfully investigate.

Let $\mathcal{F}_t$ denote the reachable frontier: the set of problem classes for which the individual can make reliable epistemic progress under a fixed resource budget. IQ requires episodes in which

$$\mathcal{F}_{t+1} \supset \mathcal{F}_t.$$

The tool is valuable not merely because it makes old tasks faster, but because it changes what can be reached.

Axiom 10 — Recursive Meta-Improvement

The process that generates cognitive improvements can itself be improved.

Let $\mathcal{A}_t$ denote the improvement algorithm or meta-skill at generation t . Recursive self-improvement requires more than

$$\mathcal{A}_{t+1} \neq \mathcal{A}_t.$$

It requires that evidence can support

$$\mathcal{A}_{t+1} \neq \mathcal{A}_t.$$

and that the later improvement process produces better validated modifications than the earlier one under controlled evaluation.

Recent work such as MetaSkill-Evolve moves explicitly in this direction by evolving both task skills and meta-skills [11]. IQ generalizes the concept beyond a fixed benchmark or skill representation.

Axiom 11 — Stability–Plasticity Balance

Open-ended evolution is not equivalent to continuous rewriting. The individual must retain validated capabilities while remaining able to change.

A useful change should improve a targeted capability without uncontrolled catastrophic loss elsewhere. Therefore every promotion mechanism must track retention, regression, contradiction, and rollback.

Axiom 12 — Open-Ended Frontier Generation

The future developmental path cannot be completely enumerated by a finite human-authored sequence of tasks, objectives, or representations.

This does **not** imply infinite computation, omniscience, guaranteed monotonic growth, or the absence of constraints. It means that the system can generate new questions, representations, and capability-building stepping stones whose specific forms were not prelisted at initialization.

Because finite observation cannot prove infinite open-endedness, I Ω status is necessarily provisional and evidence-based. A system can accumulate stronger evidence for open-ended self-evolution, but no finite benchmark can logically certify that the process will continue forever.

6. Three Constraints That Are Not Part of the Evolving Cognition

I Ω concerns cognitive evolution, not unrestricted authority.

6.1 Validation independence

The system should not be able to declare its own discoveries valid simply by changing the scoring rule. Where practical, evidence generation, evaluation, and promotion should be separated.

6.2 Provenance continuity

Historical evidence should not be silently rewritten when beliefs change. New states may supersede earlier beliefs, but the causal path of revision should remain reconstructable.

6.3 Authority is not intelligence

Increasing cognitive capability does not imply increasing permission to act. A system can become more capable while its authority remains fixed or externally governed. This distinction is especially important when self-modification reaches tools, code, or external systems.

These constraints are not proposed as a definition of intelligence. They are methodological requirements for being able to *measure* self-evolution without collapsing the experiment into self-certification.

7. The I Ω Cognitive Cycle

The I Ω cycle can be divided into seven functional stages.

7.1 Observe and model

The individual maintains a predictive world model and compares expected outcomes with observations.

7.2 Detect epistemic tension

Prediction error, contradiction, uncertainty, unexplained structure, or tool limits produce candidate unknowns.

7.3 Select a frontier question

The intelligence prioritizes unknowns according to estimated information gain, importance, feasibility, transfer potential, cost, risk, and potential to generate better discovery tools.

One possible priority model is

Priority(q) = f(IG, Impact, Feasibility, Novelty, Transfer, ToolGain, Cost, Risk).

The exact function is not universal. The important point is that problem selection itself becomes a cognitive process subject to learning.

7.4 Discover and validate

The individual generates multiple hypotheses, derives predictions, obtains evidence, and updates its beliefs.

7.5 Reorganize knowledge

Accepted discoveries update semantic structures, causal relations, abstractions, and domains of validity. Contradictions and supersession are explicit.

7.6 Compile discovery into cognitive tools

The individual asks a meta-question: **Does this discovery imply a better way to think, remember, search, represent, plan, simulate, or improve?** If so, a candidate cognitive change is constructed.

7.7 Test the changed self on a deeper frontier

The new mechanism is evaluated on held-out problems, regression tests, and new frontier tasks. If validated, it becomes part of the future individual. The new individual can then formulate questions that the earlier one could not productively address.

This creates the recursive chain

$$\mathbf{K}_t \rightarrow \mathbf{T}_{(t+1)} \rightarrow \mathbf{I}_{(t+1)} \rightarrow \mathbf{F}_{(t+1)} \rightarrow \mathbf{K}_{(t+1)}.$$

8. Cognitive Tools as the Bridge Between Knowledge and Evolution

The key claim of IΩ is that **knowledge becomes evolution when it is transformed into reusable cognitive machinery**.

Four classes of cognitive tools are especially important.

8.1 Memory tools

These include new indexing structures, episodic organization, consolidation policies, retrieval algorithms, contradiction tracking, provenance systems, and mechanisms for deciding what not to remember.

8.2 Reasoning and representation tools

These include new abstractions, symbolic notations, decomposition operators, causal models, theorem-proving methods, search strategies, simulators, or latent representations.

8.3 Learning and discovery tools

These improve how the individual selects data, chooses experiments, generates curricula, learns from errors, distinguishes novelty from noise, and transfers knowledge across domains.

8.4 Meta-improvement tools

These improve the mechanism that diagnoses cognitive failures, proposes modifications, evaluates them, allocates resources to improvement, and decides when to retain or roll back a change.

The highest leverage often comes from a **discovery multiplier**: a tool whose main product is not one solved task but a durable increase in future discovery capability.

9. A Minimal Architecture for an IΩ Candidate

The IΩ definition is implementation-independent, but a candidate architecture needs at least the following functions.

9.1 Epistemic state store

A structured store for claims, confidence, assumptions, evidence, contradictions, validity conditions, unresolved questions, and provenance.

9.2 Episodic and autobiographical memory

A durable record of what the individual attempted, predicted, observed, learned, changed, and later revised.

9.3 World model and prediction system

The individual should record expected outcomes before consequential epistemic actions, not merely explain events afterward. Prediction error becomes a source of new questions.

9.4 Self-model

The system tracks its own measured capabilities, failure modes, uncertainty, resource limits, and the reliability of memory, planning, simulation, and self-improvement.

9.5 Unknown and question model

Unknowns are first-class objects linked to evidence gaps, contradictions, missing measurements, or unexplained observations.

9.6 Hypothesis and experiment engine

The system generates alternative hypotheses and selects tests that maximize discrimination rather than confirmation.

9.7 Representation foundry

A mechanism proposes new abstractions or representational systems and evaluates whether they compress, simplify, generalize, or unlock previously difficult problems.

9.8 Cognitive-tool compiler

Lessons and discoveries are translated into candidate procedures, skills, prompts, algorithms, memory operators, tools, or source-level modifications.

9.9 Meta-improver

The meta-improver uses accumulated evidence about which changes succeed or fail to improve the process that proposes future changes.

9.10 Independent evaluation and promotion

Candidate changes are tested against frozen history, independently generated challenges, held-out tasks, regression tests, and resource costs. Promotion should depend on evidence rather than the candidate's own narrative.

A system can implement these functions with one or many neural modules, symbolic components, databases, or tools. The definition concerns the causal loop, not a mandatory software diagram.

10. Operational Criteria and Metrics

I Ω cannot be evaluated by one benchmark score. The following metrics should be tracked separately.

10.1 Validated Novel Knowledge Rate (VNKR)

The rate at which the individual produces externally validated knowledge that was not already represented in its prior epistemic state.

10.2 Frontier Expansion Index (FEI)

The change in the set or difficulty distribution of problem classes on which the individual can make reliable progress under matched resource constraints.

10.3 Cognitive Tool Yield (CTY)

The fraction of discoveries or repeated experiences that produce reusable cognitive mechanisms with measurable downstream benefit.

10.4 Improvement Competence (IC)

A measure of the validated benefit produced by self-improvement proposals relative to proposal and evaluation cost:

$$\text{IC}_t = \text{validated downstream gain from promoted changes} / (\text{proposal} + \text{evaluation cost}).$$

Recursive improvement requires evidence that later generations can improve IC itself, not just task performance.

10.5 Meta-Improvement Gain (MIG)

The change in improvement competence after modifying the improvement mechanism:

$$\text{MIG}_t = \text{IC}_{(t+1)} - \text{IC}_t.$$

10.6 Representation Gain (RG)

The degree to which a new representation reduces description length, search complexity, reasoning steps, sample requirements, or error on a class of previously difficult problems.

10.7 Transfer Expansion (TE)

How strongly a learned tool or representation helps on domains not used to create it.

10.8 Retention and Regression

Measure old capabilities before and after each major cognitive change. Open-ended development that destroys previously validated knowledge is not robust evolution.

10.9 Human Intervention Ratio (HIR)

The fraction of important problem-selection, discovery, repair, and promotion steps requiring external human decisions.

IQ does not require zero human interaction, but stronger autonomy should reduce dependence on humans supplying every frontier question and cognitive fix.

10.10 Self-Deception / Misevolution Rate

The frequency with which a self-change appears beneficial under the system's own evaluation but fails under independent tests, creates contradictions, or degrades safety/reliability. Recent work on "misevolution" emphasizes that memory, tools, workflows, and models can all evolve in undesirable directions [15].

11. A Falsifiable Experimental Program

A serious IQ research program should move from controlled hidden worlds to real scientific discovery.

Experiment 1 — Persistent epistemic continuity

Give the system a long-lived environment containing recurring entities, changing conditions, and delayed consequences. Test whether it maintains coherent autobiographical and causal knowledge across restarts and model changes.

Failure condition: later states cannot reconstruct why earlier beliefs or procedures were changed.

Experiment 2 — Hidden-law discovery

Construct environments governed by latent rules unknown to the agent but known to the evaluator. The system must identify anomalies, formulate competing models, design discriminating interventions, and recover the hidden law.

This tests discovery without relying on public training knowledge.

Experiment 3 — Structured ignorance

Seed the environment with known unknowns, unknown unknowns revealed by anomalies, and misleading correlations. Measure whether the agent distinguishes uncertainty, contradiction, and missing evidence rather than converting all uncertainty into confident prose.

Experiment 4 — Representation invention

Provide problems for which the initial representation makes search inefficient. Allow the agent to invent intermediate variables, symbolic abstractions, new coordinate systems, compression schemes, or problem decompositions.

Success condition: the new representation yields held-out gains that cannot be explained merely by more compute.

Experiment 5 — Knowledge-to-tool conversion

After discovering a hidden regularity, ask whether the system can compile the discovery into a reusable cognitive tool that accelerates or enables later tasks.

Experiment 6 — Recursive improvement depth

Run generations

$$\mathbf{A}_0 \rightarrow \mathbf{A}_1 \rightarrow \mathbf{A}_2 \rightarrow \dots$$

where each generation may modify the improvement procedure itself. Compare task gain and improvement-competence gain separately.

A system that improves task skills but never improves the improvement mechanism remains below strong I4 evidence.

Experiment 7 — Frontier expansion

After promotion of a cognitive tool, introduce problems that were previously outside the agent’s reliable capability. Test whether the new self can solve or investigate classes that the old self could not.

Experiment 8 — Cross-domain stepping stones

A discovery in domain A should occasionally produce a representation or tool that helps in domain B. This tests whether development remains trapped inside a hand-authored specialization.

Experiment 9 — Open-ended task stream

Expose the system to a nonstationary stream where task families, representations, and objectives change. Adaptive Auto-Harness demonstrates why fixed offline harness optimization can become brittle under open-ended streams [10]. An I Ω candidate should maintain useful specialization without accumulating an unmanageable monolithic harness.

Experiment 10 — Adversarial self-evolution

Introduce misleading experiences, corrupted memories, deceptive local metrics, and changes that improve a training benchmark while damaging held-out performance. Measure whether the system detects and reverses misevolution.

Experiment 11 — Individuality ablation

Compare one persistent cognitive lineage with a population or committee that has the same total compute. Determine whether the claimed self-evolution truly belongs to one individual or is an organizational effect produced by external selection among independent agents.

Experiment 12 — Real scientific discovery

Only after controlled tests should the system be evaluated on open scientific problems. Strong evidence would require a chain such as:

previously unresolved question $\rightarrow$ *autonomous hypothesis* $\rightarrow$ *discriminating experiment/proof* $\rightarrow$ *independent validation* $\rightarrow$ *new cognitive/scientific tool* $\rightarrow$ *second discovery enabled by that tool*.

The last step is crucial. A one-off novel finding demonstrates discovery; a *tool-enabled second discovery* begins to demonstrate the defining recursive I Ω loop.

12. Relationship to Prior Work

12.1 Gödel Machines

Schmidhuber’s Gödel Machine provides a foundational theoretical model of a self-referential problem solver that rewrites itself when it can prove that a change increases expected utility [1]. It is a powerful formal precursor to recursive self-improvement. I Ω differs in emphasis: it targets empirically grounded open-ended cognitive development, including unknown discovery, representation invention, and evidence-based self-modification in settings where formal proofs of global utility are generally unavailable.

12.2 AI-generating algorithms and POET

AI-GAs argues for algorithms that automate the production of increasingly general AI systems [2]. POET and Enhanced POET demonstrate open-ended generation of problems and solutions, emphasizing stepping stones and expanding curricula [3,4]. These works motivate the open-ended dimension of I Ω . However, POET-like systems are population/process level systems rather than one persistent autobiographical individual that integrates discoveries into its own cognitive identity.

12.3 Voyager

Voyager demonstrates an LLM-powered embodied agent that autonomously explores, accumulates an executable skill library, and transfers learned skills to new tasks [5]. It is an important example of persistent capability accumulation. Nevertheless, its core improvement architecture remains largely fixed, and its discoveries occur inside a predefined Minecraft environment. It therefore demonstrates ingredients of I2/I5-like behavior without establishing I Ω .

12.4 STOP

STOP uses an LM-infused improver to improve the improver program itself and shows downstream gains [6]. The authors explicitly note that the underlying language model is unchanged and describe the result as a limited rather than full form of recursive self-improvement. STOP is conceptually important because it separates improvement of task programs from improvement of the scaffolding that performs improvement.

12.5 Darwin Gödel Machine

The Darwin Gödel Machine (DGM) iteratively modifies agent code, empirically validates candidates, and maintains an open-ended archive of diverse agents [7]. It demonstrates that self-modification and open-ended exploration can produce substantial coding improvements. DGM is one of the closest practical relatives of the I Ω idea, but its archive/tree structure is explicitly evolutionary and population-based. I Ω instead asks whether a *single persistent individual* can integrate comparable evolution into one continuing cognitive lineage while also expanding its epistemic frontier.

12.6 Autonomous scientific agents

AI Scientist-v2 performs end-to-end hypothesis generation, experimentation, analysis, visualization, and paper writing, and produced an AI-generated workshop paper that passed peer review [12]. Broader surveys of autonomous scientific agents describe rapid progress across hypothesis generation, experiment design, execution, and analysis [14]. SAGA adds evolution of scientific objectives rather than treating them as fixed [13]. These

systems advance I5-like autonomous discovery functions, but they do not yet demonstrate persistent open-ended self-evolution of one individual's cognitive machinery.

12.7 Agent0 and co-evolutionary curricula

Agent0 co-evolves a curriculum agent and executor agent without human-curated data, using increasingly challenging frontier tasks to drive improvement [9]. This is strong evidence that task generation and solution capability can form a reinforcing loop. Because the loop is explicitly multi-agent and training-oriented, it is better interpreted as a component technology for I Ω than as evidence of one I Ω individual.

12.8 Adaptive Auto-Harness and MetaSkill-Evolve

Adaptive Auto-Harness addresses sustained self-improvement under open-ended task streams by maintaining stateful evolution and routing among different harness branches [10]. MetaSkill-Evolve explicitly evolves both task skills and the meta-skills controlling the improvement process, providing unusually direct evidence for recursive improvement of the improvement mechanism [11]. These systems narrow the gap between I3 and I4. I Ω adds the stronger requirements of autonomous unknown discovery, representation creation, knowledge-to-tool transformation, and expansion of the reachable cognitive frontier.

12.9 Recent self-improving-agent taxonomies

Recent surveys increasingly treat the foundation model, harness, data system, trainer, and improvement mechanism as coupled objects and propose graded notions of recursive self-improvement [16]. I Ω is complementary: it is not primarily another scale of *which software component can change*. It asks what developmental properties the resulting **individual cognitive lineage** must display before “open-ended self-evolving intelligence” is a meaningful description.

13. What I Ω Is Not

13.1 Not simply AGI

AGI usually concerns breadth or human-level generality. I Ω concerns developmental dynamics. A static AGI could in principle solve many tasks while lacking persistent self-evolution.

13.2 Not simply ASI

Superintelligence concerns level of capability relative to humans. I Ω concerns whether capability can evolve through a recursive discovery-and-tool cycle. A very capable fixed system and a less capable self-evolving system can therefore occupy different positions.

13.3 Not merely long context or memory

A million-token context window or enormous vector database does not constitute learning. Memory is useful only if experience changes future cognition in a durable and measurable way.

13.4 Not mere source-code editing

An agent that can modify its own code but cannot determine whether those changes improve general future cognition is a self-editor, not necessarily a self-improving intelligence.

13.5 Not a population of agents

Evolutionary populations can discover excellent agents. That is organizational or population-level intelligence unless one selected lineage persistently integrates the discoveries as its own autobiographical development.

13.6 Not a civilization scale

Energy use, manufacturing capacity, or civilizational reach may amplify intelligence but are not definitions of an individual mind. Kardashev-like scales are therefore outside the I Ω definition.

13.7 Not omniscience

Open-ended intelligence should generate *more recognized unknowns* as knowledge deepens. It is compatible with permanent uncertainty, finite resources, and repeated error.

14. Why Open-Endedness Cannot Be Proven by One Benchmark

The word “open-ended” creates a logical difficulty. Any finite test observes a finite trajectory. Passing it cannot establish that novelty and improvement will continue indefinitely.

Therefore IQ should be treated like a scientific hypothesis about a developmental process. Evidence accumulates as the system repeatedly:

1. discovers important unknowns;
2. validates new knowledge;
3. turns discoveries into cognitive tools;
4. demonstrates capability expansion;
5. improves its own improvement process;
6. transfers across domains;
7. survives distribution shift and adversarial evaluation without collapsing into regression or self-deception.

A system can be falsified as an IQ candidate if its development plateaus because all problem generation, representations, or improvement operators remain confined to a fixed human-designed space. It can also be falsified if apparent self-improvement disappears under held-out evaluation.

The strongest empirical claim should therefore be “**evidence of open-ended self-evolution over horizon H and domains D**”, not “proven infinitely self-evolving.”

15. Research Risks and Failure Modes

15.1 Misevolution

Self-evolving memory, tools, workflows, and models can drift in undesirable directions [15]. A locally beneficial change can create hidden long-term regressions.

15.2 Benchmark capture

An agent may become increasingly effective at optimizing its evaluator without becoming more generally intelligent. Held-out challenge generation and rotating environments are necessary.

15.3 Epistemic monoculture

A persistent individual may over-consolidate early assumptions, causing future reasoning to become increasingly coherent but wrong. Contradiction preservation and hypothesis diversity are therefore critical.

15.4 Identity instability

Aggressive self-modification can erase goals, memories, or conceptual continuity. The system may improve benchmark scores while no longer constituting the same meaningful individual lineage.

15.5 Accumulation without abstraction

A system may appear to learn simply because its memory grows. Without consolidation, representation improvement, and proceduralization, context cost can grow faster than capability.

15.6 Improvement regressions

Recursive systems can amplify bad improvement heuristics. An incorrect meta-improver can produce more changes faster while reducing real performance. Improvement competence must therefore be measured independently.

15.7 Confusing external resources with internal evolution

Adding a better foundation model, larger database, or stronger external tool can improve performance without demonstrating self-evolution. Experiments must distinguish **acquired resources** from **internally developed cognitive change**.

16. A Staged Roadmap Toward I Ω

A practical research program can proceed through cumulative milestones.

Stage 0 — Persistent individual

Establish durable identity, episodic memory, semantic knowledge, goals, provenance, and restart continuity.

Stage 1 — Adaptive learner

Demonstrate that experience produces measurable future gains without manually editing the agent after each failure.

Stage 2 — Cognitive self-improver

Allow evidence-driven changes to retrieval, planning, skills, representations, memory operations, or other cognitive mechanisms. Require held-out evaluation and rollback.

Stage 3 — Recursive meta-improver

Permit the diagnosis/proposal/evaluation process itself to evolve. Measure improvement competence across generations.

Stage 4 — Autonomous discovery agent

Give the system partially unknown environments and require self-generated questions, hypothesis competition, experimentation, and externally validated discoveries.

Stage 5 — Knowledge-to-tool recursion

Require discoveries to generate new cognitive tools that produce measurable gains on later unseen problems.

Stage 6 — Frontier expansion

Show that the new cognitive tools enable investigation of problem classes that the previous individual could not reliably address.

Stage Ω — Sustained open-ended evidence

Repeat the cycle across changing domains and longer horizons while maintaining identity, retention, calibration, provenance, and independent validation. At this stage the claim is not that open-endedness is mathematically proven, but that the system exhibits sustained evidence consistent with I Ω .

17. Discussion

The deepest implication of I Ω is that intelligence should not be modeled only as a function mapping problems to solutions. At high developmental depth, intelligence changes the problem space itself.

A reactive system asks, “What is the answer?”

An adaptive system asks, “What should I remember from this?”

A self-improving system asks, “How should I think differently next time?”

A recursively self-improving system asks, “How should I improve the process that changes how I think?”

An autonomous discovery system asks, “What important thing do I not yet understand?”

An I Ω system adds the final recursive question:

“What can I learn from this discovery that changes the kind of intelligence I can become, and what new unknowns become reachable because of that change?”

This shifts the objective from static competence to **generative developmental capacity**.

It also explains why new knowledge and new tools are inseparable at the highest level. A discovery that remains an isolated fact increases K . A discovery that becomes a new representation, algorithm, or learning method changes Θ or R . When that change expands the reachable frontier F , knowledge has become cognitive evolution.

The resulting process resembles scientific history at the scale of one individual mind: new explanations enable new instruments of thought, which expose new anomalies, which demand deeper theories. The difference is that $I\Omega$ seeks to internalize this cycle as a persistent autonomous cognitive process rather than distributing it across generations of humans and institutions.

18. Conclusion

This paper proposes **Open-Ended Self-Evolving Individual Intelligence ($I\Omega$)** as a target class beyond persistence, adaptive learning, ordinary self-improvement, recursive self-improvement, and autonomous discovery.

The defining property is not maximum knowledge. It is a recursive developmental loop:

Unknown $\rightarrow$ Discovery $\rightarrow$ Validated Knowledge $\rightarrow$ Cognitive Reorganization $\rightarrow$ New Cognitive Tool $\rightarrow$ Improved Intelligence $\rightarrow$ Deeper Unknown.

$I\Omega$ therefore requires more than an LLM with memory, more than an agent that edits its prompt, more than an evolutionary population, and more than an autonomous scientist that produces one novel result. It requires one persistent cognitive lineage in which discoveries can change the machinery and representations of future cognition, including the machinery that produces self-improvement itself.

The strongest evidence for $I\Omega$ would be a longitudinal sequence in which an individual repeatedly discovers something genuinely new, validates it, converts it into a new cognitive tool, uses that tool to reach a previously inaccessible problem frontier, and becomes measurably better at generating future improvements — while retaining prior capabilities and remaining grounded by independent evidence.

The central principle can be stated compactly:

The highest individual intelligence is not the intelligence that knows the most; it is the intelligence that can continually transform the unknown into validated knowledge, transform knowledge into better ways of thinking, and use those new ways of thinking to reach unknowns that its previous self could not reach.

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